Midterm Review

Today's Agenda

- Review RL (CST8509_RL_Intro, CST8509_02_MDP, CST8509_03_Gymnasium)
- Review Q-Learning (Lab3, Assignment1)
- Sample Written Questions

Midterm Scope

- Reinforcement Learning Fundamentals
- Basic Q-Learning with Basic "homemade" environment class
- Gymnasium custom environment, Pygame rendering
    - Q-learning with Gymnasium Cliffwalking
- Qlearning deep dive
- Stable-baselines3

Q-Learning Deep Dive

- Question: Why does our CliffWalking Example converge on the shortest path?
- Q-Learning CliffWalking animation
- Discussion
  - Why does Sarsa converge on a different path?
  - How does the initialization of the qtable affect convergence?
    - Randomized? Initialize to zero?
  - How important is setting the action-values of the terminal state to zero?

Sample Written Questions

- Questions from "Time to check your learning" slides
- Draw the diagram that represents the primary aspects of a Reinforcement Learning problem/solution with agent-environment interaction

Sample Written Questions

- Write down the q-table update portion of the q-learning algorithm in python syntax. Give a list of each variable used and its meaning.

```
qtable[state][action] = qtable[state][action] + alpha * (reward + gamma * max(qtable[next_state]) - qtable[state][action])
```

qtable: the table of action-values implementing the action-value function

state: the current state

action: the current action

alpha: step size

reward: reward received from taking action in state

gamma: discount factor

next_state: the state resulting from taking action in state

Sample Written Questions

What is gymnasium?

- **An API standard for reinforcement learning with a diverse collection of reference environments**

or

- Gymnasium is a framework for creating Reinforcement Learning environments with a standard interface such that various RL algorithms/agents can be applied to the environment in a standard way

Sample Written Questions

What is a gymnasium wrapper?

From the docs…

Wrappers are a convenient way to modify an existing environment without having to alter the underlying code directly.

In order to wrap an environment, you must first initialize a base environment. Then you can pass this environment along with (possibly optional) parameters to the wrapper's constructor.

What is stable-baselines3?

Stable-baselines3 is a set of reliable Reinforcement Learning algorithm implementations that includes features such as:

- Vectorized environments (running the algorithm on several copies of the environment at the same time)
- Callbacks (giving the programmer mechanisms to run custom code to do monitoring, auto saving, model manipulation, progress bars, etc